Summer Project

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Abstract

This project develops a simulation framework for modelling and compensating thermally induced drift in a thermo-optical medium. The optical system consists of a Gaussian input field evaluated at the spatial light modulator (SLM) plane, where a phase-only modulation pattern is applied. The modulated field then propagates through the medium, described by a time-varying complex optical operator, and the resulting intensity distribution is recorded at a downstream detector.

The thermal dynamics of the medium are governed by the heat equation, with temperature-dependent refractive-index changes introduced through the thermo-optic effect. These variations produce a spatially and temporally varying phase perturbation, which causes slow drift in the effective optical response of the system. The evolving thermal state may be captured either by a physics-informed neural network (PINN), which encodes the governing equation directly into its training objective, or by a classical numerical solver; in either case the result is a temperature field consistent with the governing physics and any available boundary, initial, and observational constraints.

Separately from this modelling component, a feedback control loop is used to update the SLM phase pattern based on the measured output intensity distribution. This control loop may be implemented using reinforcement learning, with the reward defined by the intensity at a designated target output location. In this formulation, the thermal model - whether PINN-based or numerically solved - tracks the evolving state of the medium, while the RL controller acts on optical measurements to compensate drift and recover the desired output response. The project is therefore structured around two coupled but conceptually distinct tasks: modelling the thermally driven dynamics of the medium and feedback control of the SLM phase pattern.

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1 Background Information

1.1 Optical Physics

The optical beam is modelled as a Gaussian beam propagating along the optical axis (z-axis). Working in the paraxial approximation, the complex electric field is

$$E(x,y,z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{x^2+y^2}{w(z)^2}\right) \exp\left(-ikz - i\frac{k(x^2+y^2)}{2R(z)} + i\zeta(z)\right), \tag{1}$$

where $k = 2\pi/\lambda$ is the wavenumber and the beam parameters are

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \tag{2}$$

$$R(z) = z\left(1+\left(\frac{z_R}{z}\right)^2\right), \tag{3}$$

$$\zeta(z) = \arctan\left(\frac{z}{z_R}\right), \qquad z_R = \frac{\pi w_0^2}{\lambda}. \tag{4}$$

Here w_0 is the beam waist, $w(z)$ is the beam radius at position z, $R(z)$ is the radius of curvature of the wavefronts, $\zeta(z)$ is the Gouy phase, and z_R is the Rayleigh range.

At the SLM plane $z = z_0$ the amplitude prefactor $w_0/w(z_0)$ is constant, and $\exp(-ikz_0)$ and $\exp(i\zeta(z_0))$ are *global* phases, so they may be dropped. The curvature term

$$\exp\left(-i\frac{k(x^2+y^2)}{2R(z_0)}\right)$$

is spatially varying and should be retained unless the beam is well collimated. The incident field at the SLM plane is therefore

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right)\exp\left(-i\frac{k(x^2+y^2)}{2R}\right), \tag{5}$$

where $w \equiv w(z_0)$, $R \equiv R(z_0)$, and A absorbs all constant prefactors. In the collimated limit ($R \to \infty$) this reduces to

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right). \tag{6}$$

The modulated field then propagates through thermo-optical material modelled by a linear transmission matrix $T_{\text{opt}}(t)$, and the detector measures only intensity:

$$E_{\text{out}} = T_{\text{opt}}(t)\, E_{\text{SLM}}, \qquad I(x,y) = |E_{\text{out}}(x,y)|^2. \tag{7}$$

Optical detectors measure intensity, not complex amplitude, so $I(x,y)$ is the only accessible quantity.

In summary, the incident field is the Gaussian beam evaluated at the SLM plane; spatially varying curvature phase is retained when needed; the downstream optical medium is captured by a complex transmission matrix; and the output camera measures intensity.

1.2 Thermal Physics

The optical medium is treated as a thermally perturbed phase/scattering element whose temperature field evolves on timescales much slower than the optical field. Over these slower thermal timescales, the temperature distribution $T(\mathbf{r}, t)$ is modelled by the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t), \tag{8}$$

where ρ is the density, c_p is the specific heat capacity, κ is the thermal conductivity, and $Q(\mathbf{r}, t)$ is any applied or absorbed heat source.

Temperature affects the optical response through the thermo-optic effect. To first order, the refractive index is taken to vary linearly with temperature:

$$n(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T(\mathbf{r}, t) - T_0), \tag{9}$$

where n_0 is the reference refractive index at temperature T_0, and dn/dT is the thermo-optic coefficient of the material. These refractive-index variations change the optical path length through the medium and therefore introduce a time-dependent phase perturbation

$$\Delta\phi(x, y, t) = k \int \Delta n(x, y, z, t)\, dz, \tag{10}$$

with $k = 2\pi/\lambda$ and $\Delta n = n - n_0$.

At the system level, these thermally induced phase changes are represented as a slow drift in the effective optical operator of the medium. In the transmission-matrix description used here, this is written as

$$T_{\text{opt}}(t) = T_{\text{opt},0} + \Delta T_{\text{opt}}(t), \tag{11}$$

where $T_{\text{opt},0}$ is the nominal transmission matrix and $\Delta T_{\text{opt}}(t)$ captures the time-varying perturbation caused by thermal diffusion. The key thermal-physics problem is therefore to model how heat input produces a temperature field, how that temperature field modifies refractive index, and how the resulting refractive-index changes alter the optical response seen by the detector.

1.3 SLM Behaviour

The SLM acts only in the transverse (x, y) plane and is modeled as a phase-only modulator. The applied phase pattern $\phi(x, y)$ is the control input and sets the relative phases across the aperture, thereby altering interference at the output. The modulated field is

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y) e^{i\phi(x,y)}.$$

1.4 Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) are a class of machine-learning models designed to approximate the solution of a physical system while explicitly respecting its governing equations. In a standard neural network, one typically provides input–output data pairs and trains the network to reproduce the observed mapping. A PINN extends this idea by requiring the network not only to fit data, but also to satisfy the differential equations, boundary conditions, and initial conditions that describe the underlying physics.

In the present project, the quantity of interest is the temperature field inside the thermo-optical material. The PINN represents this field by a feedforward neural network $T_\theta(\mathbf{r}, t)$, which takes a space-time point $(\mathbf{r}, t)$ as its input and produces a scalar temperature prediction as its output. Setting the input as the zeroth-layer activation, the forward pass proceeds layer by layer:

$$z^{[\ell]} = W^{[\ell]}\, a^{[\ell-1]} + b^{[\ell]}, \qquad a^{[\ell]} = \sigma\Big(z^{[\ell]}\Big), \tag{12}$$

for hidden layers $\ell = 1, \ldots, N-1$. Here $W^{[\ell]}$ is the weight matrix for layer ℓ, $b^{[\ell]}$ is the bias vector, $z^{[\ell]}$ is the pre-activation, and $a^{[\ell]}$ is the post-activation output. The final layer produces a single scalar with no output activation, interpreted directly as the predicted temperature $T_\theta(\mathbf{r}, t)$. The full set of trainable parameters is

$$\theta = \{W^{[1]}, b^{[1]}, \ldots, W^{[N]}, b^{[N]}\}, \tag{13}$$

and the role of training is to adjust θ so that the network output approximates a temperature field that is both physically admissible and consistent with any available observations.

For the thermal model considered here, the governing equation is the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t). \tag{14}$$

A PINN enforces this equation approximately by penalising violations at sampled collocation points. Substituting the network output into the left- and right-hand sides gives the PDE residual

$$\mathcal{R}_\theta(\mathbf{r}, t) = \rho c_p \frac{\partial T_\theta}{\partial t} - \nabla \cdot (\kappa \nabla T_\theta) - Q(\mathbf{r}, t). \tag{15}$$

If T_θ were the exact physical solution, this residual would vanish everywhere.

The overall training objective is a weighted sum of loss terms. Let $\{(\mathbf{r}_i^{(f)}, t_i^{(f)})\}_{i=1}^{N_f}$ denote interior collocation points used to enforce the PDE, $\{(\mathbf{r}_i^{(d)}, t_i^{(d)}, T_i^{\text{data}})\}_{i=1}^{N_d}$ the observed data points, $\{(\mathbf{r}_i^{(b)}, t_i^{(b)}, T_i^{\text{BC}})\}_{i=1}^{N_b}$ the boundary-condition samples, and $\{\mathbf{r}_i^{(0)}\}_{i=1}^{N_0}$ spatial points at the initial time $t = 0$ with prescribed initial temperature $T^{\text{init}}(\mathbf{r})$. The component losses are

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} \Big| \mathcal{R}_\theta(\mathbf{r}_i^{(f)}, t_i^{(f)}) \Big|^2, \tag{16}$$

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} \Big| T_\theta(\mathbf{r}_i^{(d)}, t_i^{(d)}) - T_i^{\text{data}} \Big|^2, \tag{17}$$

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_b} \sum_{i=1}^{N_b} \Big| T_\theta(\mathbf{r}_i^{(b)}, t_i^{(b)}) - T_i^{\text{BC}} \Big|^2, \tag{18}$$

$$\mathcal{L}_{\text{IC}} = \frac{1}{N_0} \sum_{i=1}^{N_0} \Big| T_\theta(\mathbf{r}_i^{(0)}, 0) - T^{\text{init}}(\mathbf{r}_i^{(0)}) \Big|^2, \tag{19}$$

combined into the total loss

$$\mathcal{L} = \lambda_{\text{data}}\mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}}. \tag{20}$$

Here N_f, N_d, N_b, and N_0 are the sample counts for each constraint set, and T_i^{data}, T_i^{BC}, T^{init} are the prescribed temperature values at the corresponding sample points. The scalar coefficients λ_{data}, λ_{PDE}, λ_{BC}, and λ_{IC} are weighting hyperparameters that balance the relative contribution of each term during training.

The total loss $\mathcal{L}$ is a scalar function of θ; the partial derivatives of T_θ entering $\mathcal{R}_\theta$ are obtained by automatic differentiation through the network. Training then computes the gradients

$$\frac{\partial \mathcal{L}}{\partial W^{[\ell]}}, \qquad \frac{\partial \mathcal{L}}{\partial b^{[\ell]}} \tag{21}$$

and updates the weights and biases, for example by gradient descent:

$$W^{[\ell]} \leftarrow W^{[\ell]} - \eta \frac{\partial \mathcal{L}}{\partial W^{[\ell]}}, \qquad b^{[\ell]} \leftarrow b^{[\ell]} - \eta \frac{\partial \mathcal{L}}{\partial b^{[\ell]}}. \tag{22}$$

The intermediate quantities $z^{[\ell]}$ and $a^{[\ell]}$ are not updated directly; changing $W^{[\ell]}$ and $b^{[\ell]}$ changes the next forward pass and therefore $T_\theta(\mathbf{r}, t)$, which is how the physics constraints shape the learned solution.

Once a thermally consistent approximation $T_\theta(\mathbf{r}, t)$ has been obtained, it is coupled to the optical model through the thermo-optic relation

$$n_\theta(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T_\theta(\mathbf{r}, t) - T_0), \tag{23}$$

which gives a corresponding refractive-index field. This induces a time-dependent optical phase perturbation,

$$\Delta\phi_\theta(x, y, t) = k \int (n_\theta(x, y, z, t) - n_0)\, dz, \tag{24}$$

and therefore modifies the effective optical response of the material. In the present setting, the PINN is thus used as a physics-constrained surrogate model for the thermal dynamics that drive slow drift in the optical system.

The main appeal of this approach is that it incorporates known thermal physics directly into the learning process, reducing reliance on large labelled datasets and yielding predictions that remain consistent with the governing equations even when the internal state of the medium is only partially observed through optical measurements.

1.5 Reinforcement Learning

Reinforcement learning (RL) is used here as the feedback-control layer that updates the SLM phase pattern in response to measured output intensity. The environment is the optical system with its thermally drifting medium; at each time step t the agent observes s_t derived from $I(x, y)$ (optionally augmented by the thermal model), applies an action a_t corresponding to a phase pattern $\phi(x, y)$ or its update, and receives a reward such as

$$r_t = I(x_0, y_0, t), \tag{25}$$

the intensity at a designated target location. The learned policy $\pi(a_t \mid s_t)$ therefore encodes a phase-update strategy that maintains target intensity in the presence of thermal drift.

The RL controller is distinct from the thermal model. The PINN or numerical solver predicts the evolving temperature field and thus the drift in $T_{\text{opt}}(t)$; that information may be used to shape s_t or to build a training simulator, but the control loop itself is driven by optical measurements. This separation clarifies how modelling and control couple in the overall task: the thermal model tracks drift, while RL compensates it via SLM phase updates.

2 Experimental Setup

This project implements a closed-loop optical system in which the SLM phase is adjusted to counteract slow, thermally induced drift in a thermo-optical medium. The system-level flow is defined by an input field at the SLM plane, a controllable phase pattern $\phi(x, y)$ applied by the SLM, propagation through a drifting optical operator $T_{\text{opt}}(t)$, and a measured output intensity distribution $I(x, y)$ at the detector. The control objective is to maintain or recover a desired output response, typically high intensity at a specified target location.

The thermal dynamics are modelled separately from the control law. A thermal model, implemented either as a PINN-based surrogate or a classical numerical solver, predicts the evolving temperature field and therefore the time-varying perturbation of the medium's optical response. The optical forward model then maps the SLM-modulated field through this drifting operator to produce the observable intensity distribution. This modelling component characterises how drift arises but does not decide how the SLM should be updated.

The controller closes the loop by acting on the measured output. It selects the next phase pattern based on the observed $I(x, y)$ (and, if desired, model-informed state features), with the aim of maintaining target performance despite drift. The controller may be reinforcement-learning-based, but it remains conceptually distinct from the thermal model: the thermal model describes the drifting medium, while the controller chooses the next $\phi(x, y)$ from optical feedback. In compact form, the operational chain is

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y)e^{i\phi(x,y)}, \quad E_{\text{out}} = T_{\text{opt}}(t)E_{\text{SLM}}, \quad I(x, y) = |E_{\text{out}}(x, y)|^2, \tag{26}$$

with $I(x, y)$ feeding the controller to generate the next $\phi(x, y)$.

A setup of our system is shown below in Figure 1.

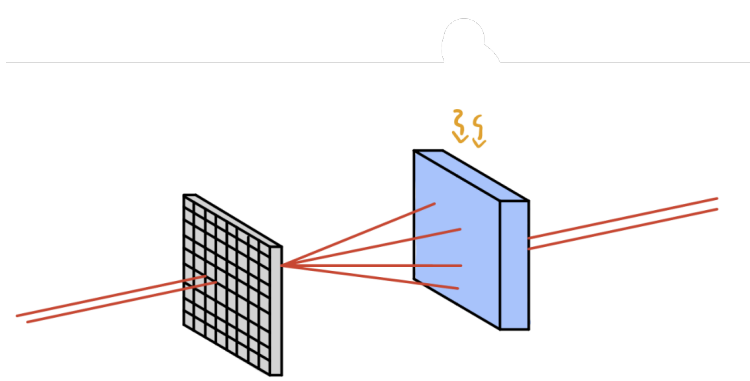

Figure 1: Schematic of the experimental setup. The laser beam is modulated by the SLM, passes through the thermo-optical material, and is detected by the camera.

The optical process is summarized as follows []:

(I) Define the incident field $E_{\text{in}}(x, y)$ at the SLM plane.

(II) Apply SLM phase modulation to obtain $E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y)e^{i\phi(x,y)}$.

(III) Propagate through the thermo-optical material using the transmission matrix $T_{\text{opt}}(t)$ to give $E_{\text{out}} = T_{\text{opt}}(t)E_{\text{SLM}}$.

(IV) Compute the output intensity $I(x, y) = |E_{\text{out}}(x, y)|^2$.

(V) Define the reinforcement-learning reward as the intensity at a selected target output location.